

GratitudeGuider: A Smart Positive Psychological Intervention System for Mental Wellbeing

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ABSTRACT

Psychological research has confirmed that positive psychological intervention methods, such as gratitude journaling, hold promising potential for promoting mental wellbeing. However, when applied in daily practices, lacking of personalized guidance has often negatively damaged its efficacy and user engagement. This paper proposes a smart system named GratitudeGuider that leverages social network data as personalized resource to provide personalized gratitude journaling service for mental wellbeing. Taking social network data as the recorder of users' real-world life, GratitudeGuider facilitates users with comprehensive stress awareness and intelligent gratitude topic recommendation aligning with their real life experience for more friendly guidance. We implemented GratitudeGuider as a lightweight mobile application and show its superiority in real user study.

CCS CONCEPTS

• Applied computing → Health care information systems.

KEYWORDS

Positive psychological intervention; gratitude journal; wellbeing; social network; deep learning

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1 INTRODUCTION

Real-world challenges such as a highly competitive society and demanding workload can easily induce anxiety and stress in daily life, negatively affects mental health. The emerging field of Positive Psychological Interventions (PPIs), is paving the way for research, applications, and advocacy resources aimed at enhancing psychological wellbeing [14]. Unlike traditional psychotherapy, which often involves costly counseling, PPIs offer simple-yet-powerful daily exercises to promote wellbeing. Gratitude journaling is a low-cost, individual-level PPI that has been proven to significantly enhance happiness and alleviate burnout, stress, and depression [1]. By writing and expressing gratitude towards kind individuals (e.g., someone who has helped you) or nice things (e.g., a beautiful day) in a daily journal, users can reimmerse themselves in these positive experiences and emotions. This practice serves as a buffer against negative moods, thereby bolstering mental health. Given the benefits to wellbeing, there has been a significant surge of interest in gratitude and its applications, particularly those developed based on internet and smartphone technologies [8].

However, the state-of-the-art gratitude intervention applications predominantly function as pure self-help tools, hardly offering guidance and lacking personalization. This results in limited efficacy, and an unsatisfactory user experience [7]. In the self-help manner, users are required to independently reflect on their past experiences and identify points of gratitude, a process that can be mentally taxing and particularly daunting for novices. Furthermore, without proper guidance, less experienced users often limit their expressions of gratitude to instances where they have received assistance from others. If no such events have occurred recently, users are prone to either skip or palter with the task [9]. While some applications attempt to stimulate gratitude thinking by providing general prosocial quotes [5], the lack of responsiveness to the user's

Table 1: Gratitude Topic Repository

Dimensions	Example Topics (part)
Effort (7)	Do volunteer work; Spend time with family
Interpersonal (9)	Be valued by others; Receive help from others
Leisure (28)	Be amused (funny videos, jokes, etc.); Decorate home; Enjoy a good night's sleep
Luck (6)	Come across good luck; Get a windfall; Pleasant surprises; Recover from illness; Recovering something lost (person or item)
Self-growth (6)	Achieve personal goals; Be productive in study/work; Complete a task; Find meanings in life

real-life context ultimately undermines the intervention's efficacy and user engagement.

To address the challenges, we propose GratitudeGuider, a novel mobile application that utilizes user's social network data as a powerful and personalized resource to serve as an easy-to-use, personalized gratitude journaling exercise. Combining previous stress detection technique from social network [12] together with professional psychological scales, our proposed application offers users with comprehensive awareness of their stress status (i.e., dynamic distributions of various categories and current level). A deep learning model is then designed to recommend appropriate gratitude topics that are closely aligned with users' real-life experiences. Structured writing guidance is also provided to help the user organize the gratitude contents. To enhance the immersive interactivity and improve user engagement, GratitudeGuide takes advantage of multi-modal elements to display gratitude journals, exploits GenAI to automatically generate images according to user's gratitude texts and allows users to save the journal as delicate pictures for online sharing.

Extending the theoretical workflow of positive psychological intervention by artificial intelligence techniques, GratitudeGuider contributes to the first smart gratitude journaling application with intelligent guidance mode instead of traditional self-help mode. Our application runs on mobile devices and enables lightweight wellbeing care in daily context. Real user study with lasting intervention experiments validates the superior efficacy compared with state-of-the-art self-help gratitude intervention services.

2 THE GRATITUDEGUIDER SYSTEM

The GratitudeGuider system comprises of three modules. Stress awareness module facilitates users with comprehensive understanding of their mental status. Intelligent gratitude journaling module guides users easily but effectively engaged in gratitude intervention. And gratitude review module enables retrospect of users' gratitude series, further strengthening positive emotions and experiences.

2.1 Comprehensive Stress Awareness

Combining professional psychological measurement and big data analysis, GratitudeGuider utilizes dual channels to comprehensively analyze users' stress level and stress categories, respectively.

2.1.1 Reliable Stress Level Assessment via Psychology Scale. The system provides default stress detection by the widely-used Perceived Stress Scale (PSS) [2] everyday when the users first launch the application. With 14 questions in the scale and each scores 1 to 5, the raw score of one's stress ranges in [14, 70]. GratitudeGuider distinguishes four stress levels, denoted as $L = \{0, 1, 2, 3\}$ corresponding to no stress, mild stress, moderate stress, and severe stress, respectively (Figure 1 ①). According to [2], assume s_u be user u 's PSS score, lv_u be the user's stress level. The mapping rule is as following: $lv_u = k$ if $s_u \in [(k+1) * 14 + 1, (k+2) * 14]$ ($k = 0, 1, 2, 3$).

2.1.2 Detailed Stress Category Distribution via Social Network. GratitudeGuider leverages users' microblog lists to supplement rich semantics of stress to help users comprehensively learn not only their stress level, but also stress category. Four common categories (i.e., academic, romantic, interpersonal and self-cognition stress) [15] are considered. And such categories were extracted from microblog with previous meta-learning technique [12]. Given date d , stress category c , and stress level lv_u , the stress category distribution is figured by multiplying lv_u with the number of microblogs expressing c on that day. Figure 1 ② visualized the dynamics of different stress categories so that users could intuitively understand the recent stress distribution.

2.2 Intelligent Gratitude Journaling Guidance

2.2.1 Topic Repository Constructing. Instead of state-of-the-art gratitude applications where users need to recall a gratitude experience in self-help manner, GratitudeGuider proposes to give smart recommendations. Thus, it is the first step to construct repository of gratitude topics. In a preliminary study, we built a Wechat gratitude group and invited 244 anonymous online users to share their grateful life events as they want. With 9-month natural tracking, a gratitude corpus of 1191 posts were collected. By exploratory factor analysis (EFA; $n=353$), totally 56 daily gratitude topics belonging to five dimensions (i.e., effort, interpersonal, leisure, luck, self-growth) were extracted and comprised the topic repository (shown in Table 1).

2.2.2 Gratitude Journaling Guidance. GratitudeGuider designs a two-step guidance to realize the smart personalization of gratitude journaling guidance to help users practice gratitude easily.

Step 1: Gratitude Topic Recommendation. Taking user's historical social network posts as their real-life recorders, we provide users with gratitude topics that are closely aligned to their real-life experiences. Given date d , assume $W = \{w_1, w_2, \dots, w_m\}$ be the recent m microblog posts of the user as input, topic dimension $d \in \{Effort, Interpersonal, Leisure, Luck, Self - growth\}$ as the output, a deep learning model shown in Figure 2 is designed and implemented to capture the user's real life expressed via microblog and recommend the most appropriate gratitude topic dimension where similar experience ever happened. Note that the learning model only outputs coarse-grained topic dimensions. To the user-end, we randomly choose k ($= 5$ in the study) fine-grained topics of the output dimension from the topic repository (Figure 1 ③) as the gratitude topic recommendation. Specifically, as psychological services are highly individualized, we train the recommendation model in a supervised manner. By collecting a sample of users'

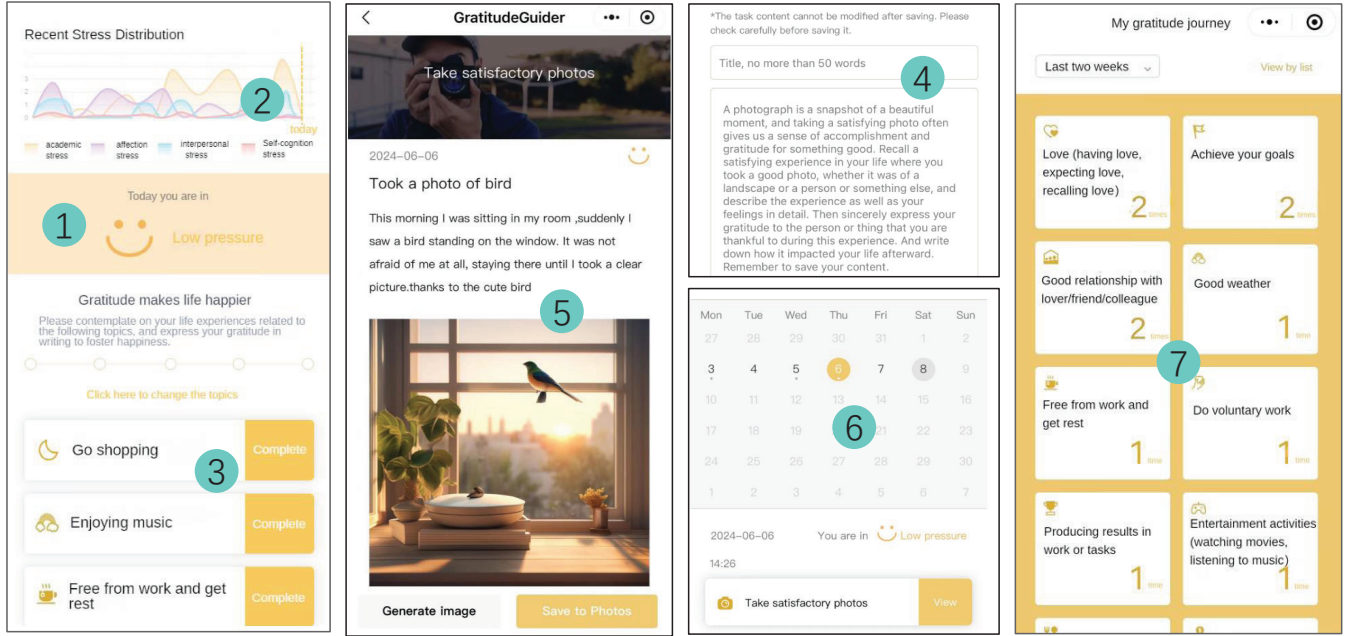


Figure 1: The overview workflow of GratitudeGuider. Users can launch the app, ② learn their detailed stress, then write gratitude journals under the guidance (③ gratitude topic recommendation and ④ structured writing guidelines), ⑤ visualize the journal, and review historical gratitude journals either ⑥ by Calender or ⑦ by topics) to alleviate stress and enhance mental wellbeing.

gratitude journals written in traditional self-help way as pilot data, we annotate the topic dimensions of the journals as training labels.

Step 2: Structured Writing Guidelines. Following principles of positive psychological studies, GratitudeGuider offers structured writing guidelines of three progressive tips to instruct users to write a gratitude journal in organized details (Figure 1 ④). First, describe the grateful experience and individual feelings at that time in detail. Then, express gratitude towards any person or things involved. Finally, reflect on whether this experience continues to impact the user’s life since then. Such sufficient details help users better immerse themselves in gratitude emotions and buffer negative status.

2.2.3 Enhancing Interactivity with GenAI. To further improve the immersive experience of gratitude journaling and enhance user engagement, we involve multi-modal elements to display the gratitude journals. Beside the users’ stress states and written contents, GratitudeGuider leverages GenAI to generate images according to journal content (as in Figure 1 ⑤). With the access to Qwen¹, keywords of the gratitude journal contents are extracted and utilized as image generator prompt. Users can then save the exquisite gratitude journals as photos, send the pieces to whom they are grateful to, or share on social media, etc.

2.3 Gratitude Review

In addition to writing gratitude journals, reviewing historical journals will also awaken positive emotions and strengthen wellbeing. GratitudeGuider constructs two alternative indexes to maintain user’s gratitude journals. One is by calendar, where each date corresponds with a list of user’s gratitude journals (Figure 1 ⑥). The

¹one of the leading generative AI developed by Aliyun in China, <https://tongyi.aliyun.com/>

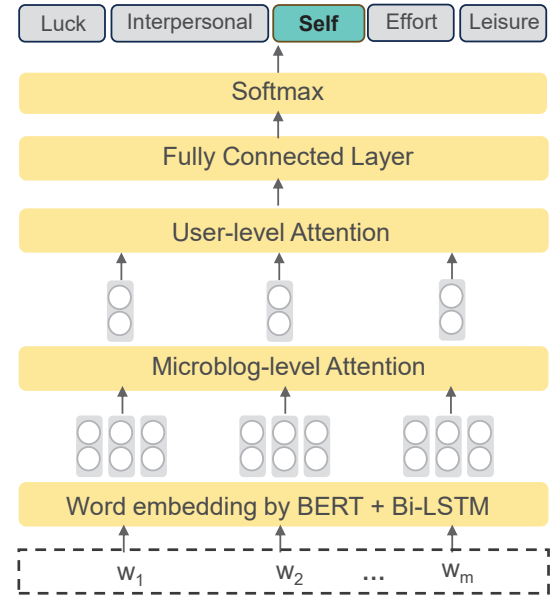


Figure 2: The deep learning model for gratitude topic recommendation.

other one is by gratitude topics, where the key becomes the recommended topics (Figure 1 ⑦). Both indexes maintain all the details of the journal, such as the date, user’s stress status, journal contents, etc. Users are able to choose either index to review his/her historical gratitude journals during the given time interval (e.g., last week, last month, etc.).

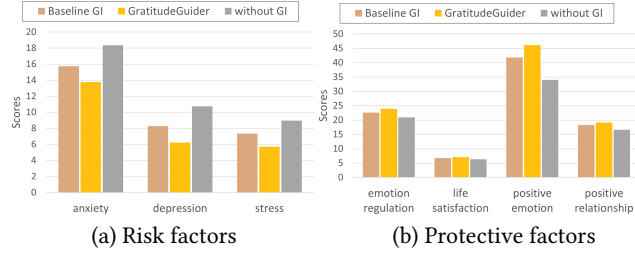


Figure 3: The superiority of GratitudeGuider in user study.

3 DEMONSTRATION

Users can utilize GratitudeGuider to understand their daily stress levels, write gratitude journals under personalized guidance to alleviate negative emotions, and review their gratitude journal repository to enhance wellbeing. The workflow is shown in Figure 1.

Daily Stress Awareness: When first logging in, the user binds his/her microblog account with GratitudeGuider. Every day when the user first enter the system, s/he fills out the brief psychological scale provided by the system to report his/her current mental status. Then his/her comprehensive psychological stress will be visualized. The fine-grained stress dynamics of academic, affection, interpersonal and self-cognition stress are analyzed from social network data (Figure 1 ②) and today's stress level is from the scale measurement (Figure 1 ①).

Gratitude Journaling Guidance: To help alleviate stress, the system guides the user in gratitude journaling exercises. According to the user's recent daily life experiences revealed from his/her social network history, GratitudeGuider intelligently suggests k ($= 5$ in the study) gratitude topics to help recall corresponding experiences worth being grateful for (Figure 1 ③). Note that the "click here to change the topics" button allows the user to change the recommended gratitude topics. When clicking the "complete" button of a selected topic tab, the user enters the writing interface and the system provides users with structured guidance on how to write a gratitude journal in detail (Figure 1 ④). When finishing the writing part, the user can click the "Generate image" button to automatically attach correlated image and decorate the journal (Figure 1 ⑤). S/he is also able to save the decorated gratitude journals as delicate photos to their phones, which is convenient for him/her to send them to whom s/he expresses gratitude to or share on social media.

Gratitude Review: The user can freely review his/her personal gratitude journal repository to reimmerge into the positive emotions. Entering "My gratitude journey" page, the user selects "view by calendar" or "view by topics" in the top right corner. In the former mode, the user obtains his/her journal lists with details such as stress status, gratitude topics and timestamps on selected date (Figure 1 ⑥). While in the latter mode, after selecting the time interval for review, e.g., the last two weeks, the user is then exposed with all the gratitude journals written at that time, indexed by different topics and frequency statistics (Figure 1 ⑦). s/he can read the journals by clicking into the tabs in both modes.

4 USER STUDY

We conducted a real user study to validate the effectiveness of GratitudeGuider, which was approved by ethical committee of Wuhan University. A total of 126 students from Wuhan University (female 66%, male 34%; with the average age of 19.6 ± 1.796 years

old) were recruited in a 2-week gratitude intervention experiment. The participants were randomly assigned to three experimental conditions. Besides utilizing GratitudeGuider as the gratitude intervention tool, the other two comparative conditions used traditional gratitude intervention approach (i.e., writing gratitude journals in self-help manner without gratitude topic recommendation) as baseline GI, and without gratitude intervention as control group, respectively. During the 2-week experiment, the participants were provided with corresponding interface everyday, monitored their stress, and required to wrote at least one gratitude journal except the control group. After the intervention, all participants filled out questionnaires to evaluate the effectiveness of the intervention². Two kinds of mental wellbeing factors were measured by professional psychology scales, risk factors (i.e., anxiety symptoms [11], depression symptoms [10] as well as perceived stress [2]) and protective factors (i.e., emotion regulation [6], positive emotions [13], life satisfaction [3], and positive relationships with others [4]).

Figure 3 compares the effectiveness of different gratitude interventions. Compared with baseline GI, GratitudeGuider significantly decreased anxiety symptoms, depression symptoms and perceived stress by 12.6%, 24.5%, and 22.3%, respectively (Figure 3(a)). Meanwhile, GratitudeGuider also increased users' positive emotion, life satisfaction, emotion regulation level and positive relationships with others by 10.2%, 5.0%, 5.7%, 4.8% than baseline GI, respectively (Figure 3(b)). In summary, with smart gratitude topic recommendation, GratitudeGuider offers more friendly and easy-to-use gratitude guidance, outperforming both baseline GI and control group, by dramatically relieving the risk factors and strengthening the protective factors of mental wellbeing.

5 CONCLUSION

We proposed GratitudeGuider, a novel mobile application that integrates positive psychological intervention theory (i.e., gratitude journaling) with deep learning approaches to facilitate intelligent psychological service for mental wellbeing. Leveraging social network data as users' real-life recorders, GratitudeGuider offers personalized gratitude topic recommendation and structured writing guidance to help stress relief and mood lift, facilitating an easy-to-use yet professional mental health service.

There are some interesting topics to be explored in the future. Instead of PSS scale, multi-source data from smart bracelets, like sleep quality, heart rate, blood oxygen etc., could be involved for automatic stress comprehension to alleviate user's interaction burden. An extension of an ecosystem that allows multiple users to interact and share contents is also worthwhile to explore, so that collaborative recommendation of gratitude topics could work for the users without posts in social network.

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²Note that measurements by the psychological scales showed for all the participants, there were no significant difference in the mental health related factors before the intervention experiment.

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